Project Data

GeoMet: spatial copper measurements

Hoffimann, J., et al. (2022). *GeoMet dataset*, version 4. DOI: 10.5281/zenodo.7051975. CC BY 4.0.

The source `drillholes.csv` contains chemical analyses and the Cartesian coordinates of drill-core sample centroids. It is the **obfuscated public version of measurements from a real deposit**. The source does not describe the precise obfuscation procedure in its public data-availability statement. Treat the released coordinates as local Cartesian positions, not geographic latitude and longitude.

`spatial_copper.csv` preserves all 2,000 source rows and the selected numerical values without rounding, clipping, imputation or transformation. We selected and renamed columns and added a zero-based source-row identifier.

Column	Meaning	Unit / role
sample_id	Source row index, excluding the header	Identifier only
hole_id	Source HOLEID; one of 119 drill holes	Grouping only
x, y, z	Released sample-centroid coordinates	Metres; model inputs
copper_ppm	Source Cu ppm concentration	ppm; response

Fixed spatial split

`spatial_split.npz` contains zero-based CSV row indices `train_idx`, `val_idx` and `test_idx`. Read source hole IDs as strings, obtain their lexically sorted unique values, and shuffle once with NumPy `default_rng(3230)`. Assign the first 71 holes to training, the next 24 to validation, and the remaining 24 to test. Preserve source-row order within each set. The indices are supplied; do not recreate a different split.

Set	Rows	Drill holes
Training	1,164	71
Validation	412	24
Test	424	24

No hole occurs in more than one set. This evaluates held-out drill holes, not separation into distant geographic regions. The course package contains measured responses at sample locations only. Prediction grids and slices do not have known truth.

Documentation: author's Mineral deposits chapter; research paper and data availability.

MNIST: a fixed small subset

MNIST by Yann LeCun and Corinna Cortes, obtained from the official Keras NPZ distribution. The Keras documentation identifies the dataset license as CC BY-SA 3.0. This selected subset retains that license and attribution.

`mnist_small.npz` contains unchanged 28 by 28 grayscale images and their original digit labels. Selection and row reordering are the only changes. Pixels and labels are stored as uint8. The archive is about 1.3 MB; students do not download or train on the full MNIST collection.

Arrays	Images	Per digit	Official source
x_train, y_train	5,000	500	Training split
x_val, y_val	1,000	100	Training split; disjoint from course training
x_test, y_test	2,000	200	Test split

Selection procedure

Initialize an independent `default_rng(3230)`. For digits 0 through 9 in order, permute matching official training indices: take the first 500 for course training and the next 100 for validation. Then, for digits 0 through 9, permute matching official test indices and take the first 200. Concatenate each set in digit order. Using the same RNG state, shuffle the training indices, then validation indices, then test indices.

`source_train_indices` and `source_val_indices` index the original official training arrays; `source_test_indices` indexes the separate official test arrays. Identical index numbers across these two source arrays are not the same image.

In the notebook, divide pixel values by 255 and flatten them. For 5-versus-rest, relabel digit 5 as 1 and all other digits as 0 without changing image membership. The binary positive counts are 500 / 100 / 200, so an always-negative rule has 90% accuracy and zero recall in every course split.

Provenance and integrity

`data_manifest.json` records source URLs and SHA-256 hashes, source licenses, field definitions, exact selection steps, and checksums of the three data files. It also records the GeoMet modifications and the MNIST subset selection. Retain this documentation when redistributing the data.

Official MNIST NPZ source: TensorFlow / Keras distribution.

Source SHA-256 (MNIST):

```
731c5ac602752760c8e48fbffcf8c3b8
50d9dc2a2aedcf2cc48468fc17b673d1
```